

Erdős Problem #836

EP-836	L1 - Tractable	Partiellement résolu	Number Theory
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Énoncé original (anglais, reproduction mot pour mot)

Let $r \geq 2$ and G be a r -uniform hypergraph with chromatic number $\chi(G) = 3$ (that is, there is a 3-colouring of the vertices of G such that no edge is monochromatic).

Suppose any two edges of G have a non-empty intersection. Must G contain $O(r^2)$ many vertices? Must there be two edges which meet in $\gg r$ many vertices?

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